GATE ECE 2027 -- AUGUST TO DECEMBER MASTER BLUEPRINT

Target: IIIT Bangalore (M.Tech ECE)
Score Target: GATE Score >= 650 | Raw Marks ~68-74/100
Timeline: Aug 15 -> Dec 31 (18 Weeks Strict)
Daily Rhythm: 6.5-7.0h Deep Work + 2.0-2.5h Gym/Recovery

OUTPUT 1: THE 18-WEEK MASTER CALENDAR (AUG 15 -> DEC 31)

Wk	Dates	Subject A (Core Hard)	Subject B (Support)	PYQs	Saturday Test	Net Target
1	Aug 15 - Aug 21	Math: Linear Algebra	GA: Verbal Ability	85	Subject Test 1: Linear Algebra (30M)	8 Marks
2	Aug 22 - Aug 28	Math: Calculus & Multi	GA: Quant 1 (Arithmetic)	85	Subject Test 2: Calculus + GA (30M)	15 Marks
3	Aug 29 - Sep 04	Math: ODEs & Vector Calc	GA: Quant 2 (P&C / Prob)	90	Subject Test 3: ODEs + Vector (30M)	22 Marks
4	**Sep 05 - Sep 11**	**Math: Probability & Complex**	**GA: Spatial & Data Interp**	**100**	**STAGE 1 SIMULATION TEST (2[illegible]**	**[illegible]M (STAGE 1 DONE)**
5	Sep 12 - Sep 18	Networks: Laws & Theorems	Digital: Boolean & MUX	90	Subject Test 4: Networks 1 + Digital 1	33 Marks
6	Sep 19 - Sep 25	Networks: Transients & AC	Digital: Counters & FFs	95	Subject Test 5: Transients + Counters	39 Marks
7	**Sep 26 - Oct 02**	**Networks: 2-Port & Reson**	**Digital: FSM & CMOS**	**90**	**SECTIONAL: Networks + Digital (60M)**	**[illegible] (STAGE 2A DONE)**
8	Oct 03 - Oct 09	Signals: CT Signals & LTI	EDC: Semiconductor Phys	85	Subject Test 6: CT Signals + EDC 1	47 Marks
9	Oct 10 - Oct 16	Signals: CTFT & Fourier	EDC: P-N Junction Diode	90	Subject Test 7: Fourier + Diode	50 Marks
10	Oct 17 - Oct 23	Signals: Laplace & Sampling	EDC: MOS Cap & MOSFET	90	Subject Test 8: Laplace + MOSFET	53 Marks
11	**Oct 24 - Oct 30**	**Signals: DTFT & Z-Transform**	**EDC: BJT Physics & Model**	**95**	**SECTIONAL: Signals + EDC (60M)**	**[illegible]M (STAGE 2B DONE)**
12	Oct 31 - Nov 06	Analog: Op-Amps & Comparators	Control: Transfer Func & SFG	85	Subject Test 9: Op-Amps + SFG	59 Marks
13	Nov 07 - Nov 13	Analog: BJT/MOSFET Amps	Control: 2nd Order Time Resp	90	Subject Test 10: Amps + Time Resp	62 Marks
14	Nov 14 - Nov 20	Analog: Feedback & Clippers	Control: Routh & Root Locus	90	Subject Test 11: Feedback + Root Loc	65 Marks
15	**Nov 21 - Nov 27**	**Analog: Oscillators & 555**	**Control: Bode & State Space**	**100**	**SECTIONAL: Analog + Control (60M)**	**[illegible]M (STAGE 3 DONE)**
16	Nov 28 - Dec 04	Comms: Random Proc & Noise	EMTL: Transmission Lines	95	Subject Test 12: Random Proc + T-Lin	71 Marks
17	Dec 05 - Dec 11	Comms: AM, FM & Pulse (PCM)	EMTL: Maxwell & Plane Waves	95	Subject Test 13: Analog Comms + Wa	74 Marks
18	**Dec 12 - Dec 18**	**Comms: Digital Mod & Info**	**EMTL: Waveguides & Antennas**	**100**	**SECTIONAL: Comms + EMTL (60M)**	**[illegible]M (STAGE 4 DONE)**
19	**Dec 19 - Dec 31**	**GRAND SYLLABUS REVISION**	**PYQ PASS 2 SPEED MARATHON**	**150**	**FULL CBT DIAGNOSTIC MOCKS**	**100% SYLLABUS DONE**

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OUTPUT 2: STAGE-BY-STAGE GRANULAR BREAKDOWN & TIER 1 TOPICS

STAGE 1: The Unshakable Foundation (Weeks 1-4 | 28M Locked) *Engg Math (13M) + General Aptitude (15M)*

W1: Linear Algebra (Eigenvalues, Cayley-Hamilton, Rank-Nullity, Ax=b) | W2: Calculus (Limits, Max-Min rt-s^2, Taylor, Vector Theorems) | W3: ODEs & Prob (Linear ODEs, Bayes, Uniform/Normal/Exp) | W4: Complex & Transforms (C-R, Residues).
Tier 1: Eigenvalues/Vectors, Cayley-Hamilton, Max-Min, Vector Theorems, Bayes' Theorem, Residue Theorem.

STAGE 2A: Circuits & Deterministic Logic (Weeks 5-7 | +14M | 44M Cumulative) *Network Theory (7M) + Digital Circuits (9M)*

W5: Network Laws/Theorems (Thevenin with dep sources, Max Power) + Digital (K-Maps, MUX, Decoders) | W6: Transients & Resonance (1st order v(t) formula, Series/Parallel Q) + Digital (Counters, FFs) | W7: Two-Port (Z,Y,h,ABCD, Reciprocity) + Digital (Moore/Mealy FSM, Setup/Hold, CMOS).
Tier 1: Thevenin with Dep Sources, Universal Transient Formula, Modulo-N Counters, Setup/Hold Slack, Two-Port Reciprocity.

STAGE 2B: Transforms & Solid-State Physics (Weeks 8-11 | +13M | 56M Cumulative) *Signals & Systems (9M) + Electronic Devices EDC (7M)*

W8: CT Signals & LTI (Graphical Convolution, BIBO) + EDC (Semiconductor physics, Fermi level, Drift/Diff) | W9: Fourier (CTFT properties, Parseval) + EDC (P-N diode Vbi, Depletion width, Cj/Cd) | W10: Laplace & Sampling (Laplace ROC, Nyquist) + EDC (MOS Cap, MOSFET ID) | W11: DT & Z-Transform + EDC (BJT Early effect, gm/rpi).
Tier 1: Graphical Convolution, CTFT Properties, Nyquist Rate, MOSFET Triode/Sat ID, P-N Diode Cj.

STAGE 3: Amplifiers, Control & Synthesis (Weeks 12-15 | +12M | 68M Cumulative) *Analog Circuits (8M) + Control Systems (7M)*

W12: Op-Amps (Virtual ground, Integrator, Schmitt trigger) + Control (Transfer Functions, SFG Mason) | W13: Transistor Amplifiers (CE/CS gain) + Control (2nd order Mp, ts, ess) | W14: Feedback Amplifiers + Clippers + Control (Routh, Root Locus) | W15: Oscillators + 555 + Control (Bode Plots GM/PM, Nyquist N=P-Z, State Space e^At).
Tier 1: Op-Amp Virtual Ground, CS/CE Small-Signal Gain, Bode Plot Transfer Function Extraction, Root Locus, 2nd Order Mp.

STAGE 4: High-Weightage Advanced Frontier (Weeks 16-18 | +15M | Net ~72-75M) *Communications (13M) + Electromagnetics EMTL (8M)*

W16: Random Processes & Noise (WSS, Autocorr, PSD S_XX, LTI noise) + EMTL (Transmission lines Zin, VSWR, Gamma) | W17: Analog & Pulse Comms (AM power, FM Carson BW, PCM SNRq) + EMTL (Maxwell, Plane waves, Skin depth, Poynting) | W18: Digital Mod & Info (BPSK/QPSK constellation, BER, Matched Filter, Shannon Capacity) + EMTL (Waveguides TE10, Antennas).
Tier 1: WSS Noise through LTI Filters, Transmission Line Zin at lambda/4, BPSK/QPSK BER, Shannon Channel Capacity, Waveguides TE10.

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OUTPUT 3: HIGH-YIELD RESOURCE & FORMULA SHORTCUTS STACK

Subject	Textbook	Video Series	Must-Have Formula Shortcuts
Engg. Math	B.S. Grewal	Dr. S.K. Gupta (NPTEL)	Sum lambda = Tr(A), Prod lambda = det(A); Ax=b unique if rank(A)=rank(A|b)=n; Bayes Theorem.
Network Theory	Hayt & Kemmerly	Prof. T.K. Basu (IIT KGP)	v(t) = v(inf) + [v(0+) - v(inf)]e^(-t/tau); Series Res Q = (1/R)sqrt(L/C); Thevenin Rth = Voc/Isc.
Digital Circuits	Morris Mano	Neso Academy (YouTube)	MUX 2^n -> 1 implements any n+1 var logic; Ripple counter fmax = 1/(N*tpd); Setup: Tclk >= tcq +
Signals & Systems	Oppenheim & Willsky	Prof. S.C. Dutta Roy (IITD)	Convolution length L = L1 + L2 - 1; CTFT rect(t/tau) <-> tau*sinc(f*tau); Laplace stability: poles in L
Electronic Devices	Streetman & Banerjee	Prof. Amitava Das Gupta	Vbi = Vt*ln(NA*ND/ni^2); MOSFET Sat: VDS >= VGS-Vth and ID = 0.5*mu*Cox*(W/L)(VGS-Vth)^2
Analog Circuits	Sedra & Smith	Prof. Shanthi Pavan (IITM)	Op-Amp Virtual Ground V+ = V-; CS/CE Gain Av = -gm*RL; Schmitt Trigger Hysteresis VH = (R1/(
Control Systems	Ogata	Prof. C.S. Shankar Ram	Mp = e^(-pi*zeta / sqrt(1-zeta^2)) * 100%; ts = 4/(zeta*wn); Mason Gain: T = (Sum Pk*Delta_k)/De
Communications	Simon Haykin	Prof. Amitabha Mukherjee	Pt = Pc(1+mu^2/2); Carson: BW = 2(Delta_f + fm); PCM (SNR)q = 1.76 + 6n dB; BPSK Pe = Q(sq
Electromagnetics	Hayt & Buck	Prof. Sachidananda (IISc)	Gamma = (ZL-Z0)/(ZL+Z0); VSWR = (1+|Gamma|)/(1-|Gamma|); Zin(lambda/4) = Z0^2/ZL; Skin de
General Aptitude	R.S. Aggarwal	CareerRide (YouTube)	Speed = Dist/Time; Relative speed (same: S1-S2, opp: S1+S2); Work: 1/T = 1/Ta + 1/Tb; Bayes &

OUTPUT 4: EMERGENCY SYLLABUS TRIAGE MATRIX (WHEN BEHIND SCHEDULE)

Subject	DROP THESE FIRST (Low ROI / High Friction)	PROTECT THESE AT ALL COSTS (High Yield)
EMTL	Antennas, Electrostatics volume integrals	**Transmission Lines (Zin, VSWR, Matching) & Waves**
Analog Circuits	555 Timers, Power Amplifiers, Complex Oscillators	**Op-Amps, CE/CS Small-Signal Gain, Feedback Topologies**
Control Systems	Non-canonical State Space realization proofs	**Bode Plot (GM/PM), Root Locus, 2nd Order Mp, Routh Array**
Electronic Devices	Hall Effect, Solar Cells, Lasers, Tunnel Diodes	**MOSFET Triode/Saturation Equations, PN Diode Vbi & Capacitance**
Communications	Superheterodyne receiver hardware circuits	**Random Processes through LTI, BPSK/QPSK BER, AM/FM Bandwidth**
Digital Circuits	Obscure ECL/TTL transistor gates	**Multiplexers (MUX), Modulo Counters, Setup/Hold Timing Slack**
Engg. Mathematics	Numerical Methods, Complex PDE separation derivations	**Linear Algebra (Eigenvalues/Rank), Calculus, Bayes' Theorem**

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OUTPUT 5: DECEMBER 31 FINISH-LINE AUDIT & JANUARY MOCK MACHINE

December 31 Verification Checklist (Certifying 100% Complete Syllabus):

[ ] 10/10 Subjects 100% Complete: All concept lectures viewed and core derivations understood.
[ ] Pass 1 of PYQs Complete: 1,500+ past GATE problems solved and tagged (Green / Yellow / Red Star).
[ ] 40-Page Master Short Notes Binder: Handwritten, high-density formula sheets complete for all 10 subjects.
[ ] Master Error Log: 300+ categorized mistakes analyzed with their anti-mistake directives.
[ ] 13 Subject Tests + 4 Sectional Tests Completed and Analyzed.
[ ] Zero Unfinished Backlog: Any dropped Tier 3 topics are noted and accepted.

The 4-Week January Mock Test Calendar (Peak Performance Training):

January Week	Mock Tests	Focus & Strategic Objective	Benchmark
Week 1 (Jan 01 - Jan 07)	Mocks #3, #4, #5 (Alt Days)	Time management & eliminate Category B silly errors	55 - 60 Marks
Week 2 (Jan 08 - Jan 14)	Mocks #6, #7, #8, #9 (Alt Days)	50-30-20 exam pacing & negative mark discipline	60 - 65 Marks
Week 3 (Jan 15 - Jan 21)	Mocks #10, #11, #12, #13 (Strict CB	Circadian 09:30 AM-12:30 PM peak alertness calibration	65 - 70 Marks
Week 4 (Jan 22 - Jan 28)	Mocks #14, #15 (Dress Rehearsals)	Error log final speed-run & formula recall	68 - 74 Marks
Final 7 Days (Jan 29+)	ZERO Full Mocks. Speed revision.	Short notes review, physical deload, sleep 8h+ mandatory	Peak Calm & Flow

THE 3-HOUR MOCK CBT SIMULATION & 2-HOUR 5-STEP ANALYSIS PROTOCOL

- 09:30 AM - 12:30 PM: Full Mock CBT Simulation (Strictly timed, desktop interface, virtual calculator only).
- 01:30 PM - 03:30 PM: 5-Step Mock Dissection: 1. Score Breakdown -> 2. Categorize A/B/C/D -> 3. Error Log -> 4. Rework -> 5. Fix Leak.
- Final 7 Days (Feb 01+): ZERO Full Mocks. Speed-read 40-page Short Notes binder, deload workout, sleep 8h+ mandatory.